SHETH L.U.J AND SIR M.V. COLLEGE
SUBJECT NAME: Cyber **& Information Security**
Practical: 3 Message Authentication Codes

AIM: Implement algorithms to generate and verify message authentication codes (MACs) for ensuring data integrity and authenticity

Code :

```
import hmac
import hashlib

# User input
secret_key = input("Enter the secret key: ").encode()
message = input("Enter the message: ").encode()

# Generate MAC
mac = hmac.new(secret_key, message, hashlib.sha256).hexdigest()

print("\nGenerated MAC:", mac)

# Verification
received_message = input("\nEnter the message for verification: ").encode()

# Generate MAC for the received message
received_mac = hmac.new(secret_key, received_message, hashlib.sha256).hexdigest()

print("\nVerifying Message...")

if hmac.compare_digest(mac, received_mac):
    print("Authentication Successful!")
    print("Message Integrity Verified.")
else:
    print("Authentication Failed!")
    print("Message has been modified.")
```

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Output :

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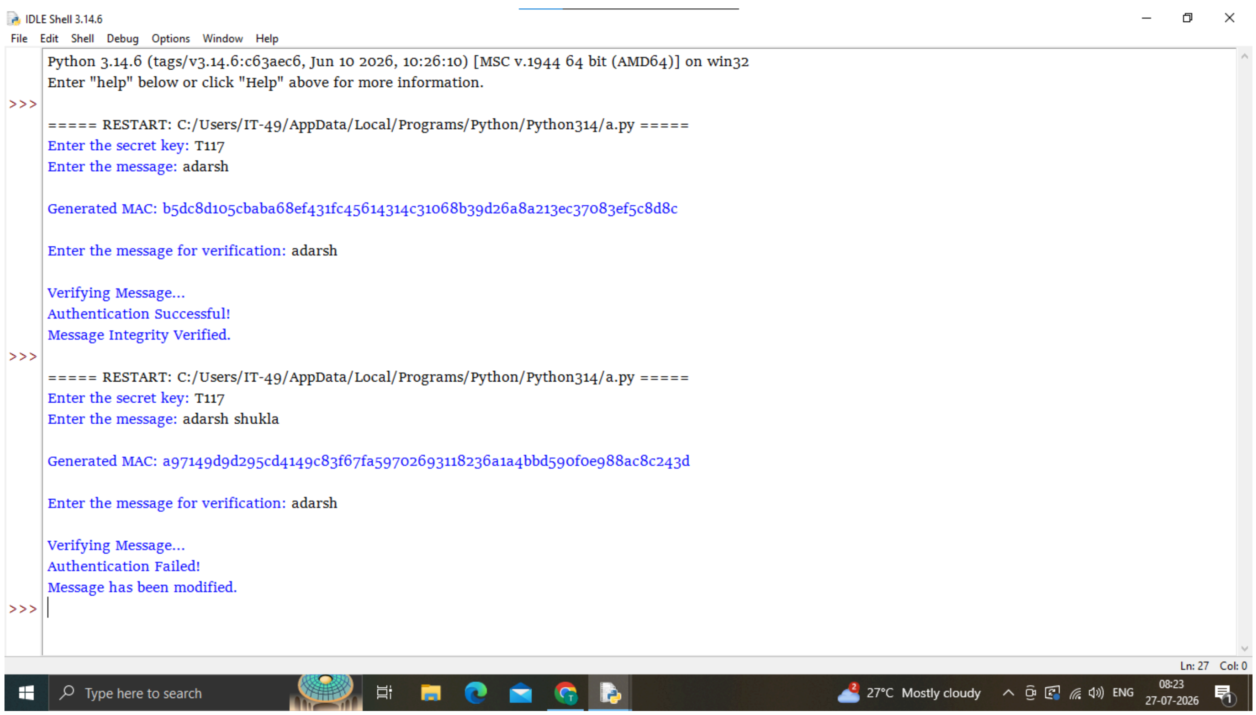


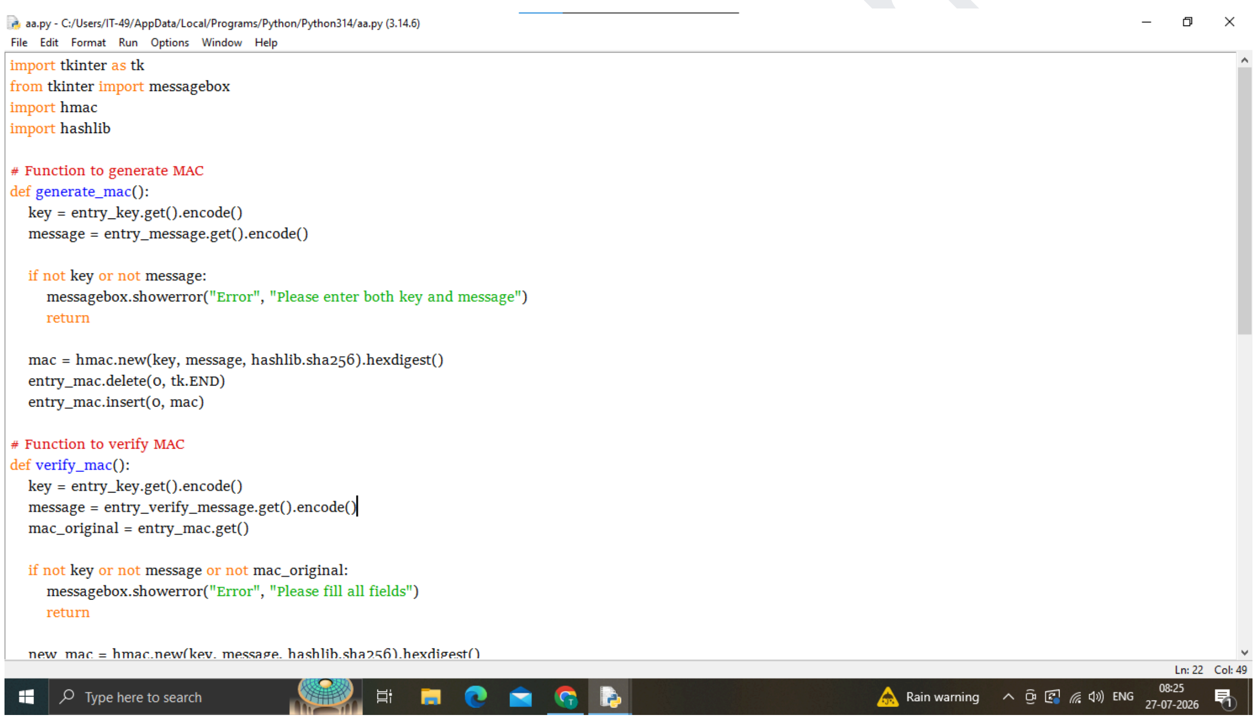
Gui :

Code :

```
import tkinter as tk
from tkinter import messagebox
import hmac
import hashlib

# Function to generate MAC
def generate_mac():
    key = entry_key.get().encode()
    message = entry_message.get().encode()

    if not key or not message:
        messagebox.showerror("Error", "Please enter both key and message")
```

```
        return

    mac = hmac.new(key, message, hashlib.sha256).hexdigest()
    entry_mac.delete(0, tk.END)
    entry_mac.insert(0, mac)

# Function to verify MAC
def verify_mac():
    key = entry_key.get().encode()
    message = entry_verify_message.get().encode()
    mac_original = entry_mac.get()

    if not key or not message or not mac_original:
        messagebox.showerror("Error", "Please fill all fields")
        return

    new_mac = hmac.new(key, message, hashlib.sha256).hexdigest()

    if hmac.compare_digest(mac_original, new_mac):
        messagebox.showinfo("Result", "✅ Authentication Successful!\nMessage Integrity Verified.")
    else:
        messagebox.showerror("Result", "❌ Authentication Failed!\nMessage has been modified.")

# GUI Window
root = tk.Tk()
root.title("MAC Generator & Verifier")
root.geometry("500x400")

# Labels & Entries
tk.Label(root, text="Secret Key").pack()
entry_key = tk.Entry(root, width=50)
entry_key.pack()

tk.Label(root, text="Message (Generate MAC)").pack()
entry_message = tk.Entry(root, width=50)
entry_message.pack()

tk.Button(root, text="Generate MAC", command=generate_mac, bg="lightblue").pack(pady=5)

tk.Label(root, text="Generated MAC").pack()
```

```
entry_mac = tk.Entry(root, width=50)
entry_mac.pack()

tk.Label(root, text="Message (Verify)").pack()
entry_verify_message = tk.Entry(root, width=50)
entry_verify_message.pack()

tk.Button(root, text="Verify MAC", command=verify_mac, bg="lightgreen").pack(pady=10)

# Run GUI
root.mainloop()
```



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Practical: 3 Message Authentication Codes

aa.py - C:/Users/IT-49/AppData/Local/Programs/Python/Python314/aa.py (3.14.6)

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```python
    if hmac.compare_digest(mac_original, new_mac):
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```

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Output :

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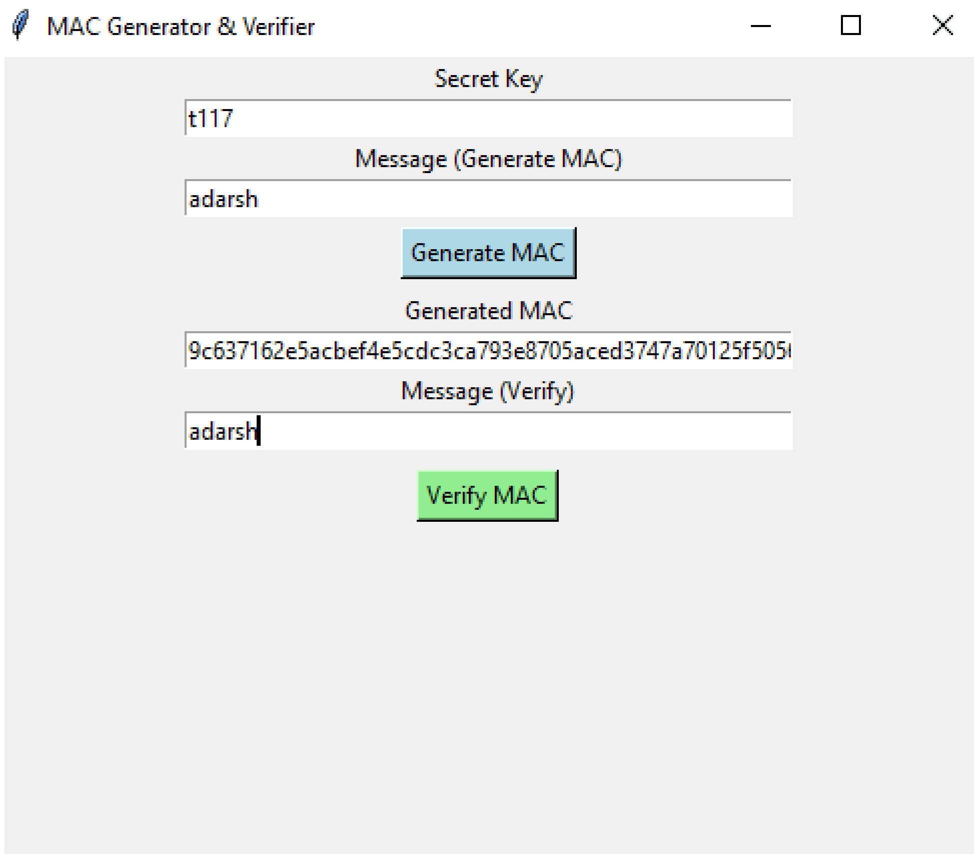
MAC Generator & Verifier
Secret Key
t117
Message (Generate MAC)
adarsh
Generate MAC
Generated MAC
9c637162e5acbef4e5cdc3ca793e8705aced3747a70125f505
Message (Verify)
adarsh
Verify MAC

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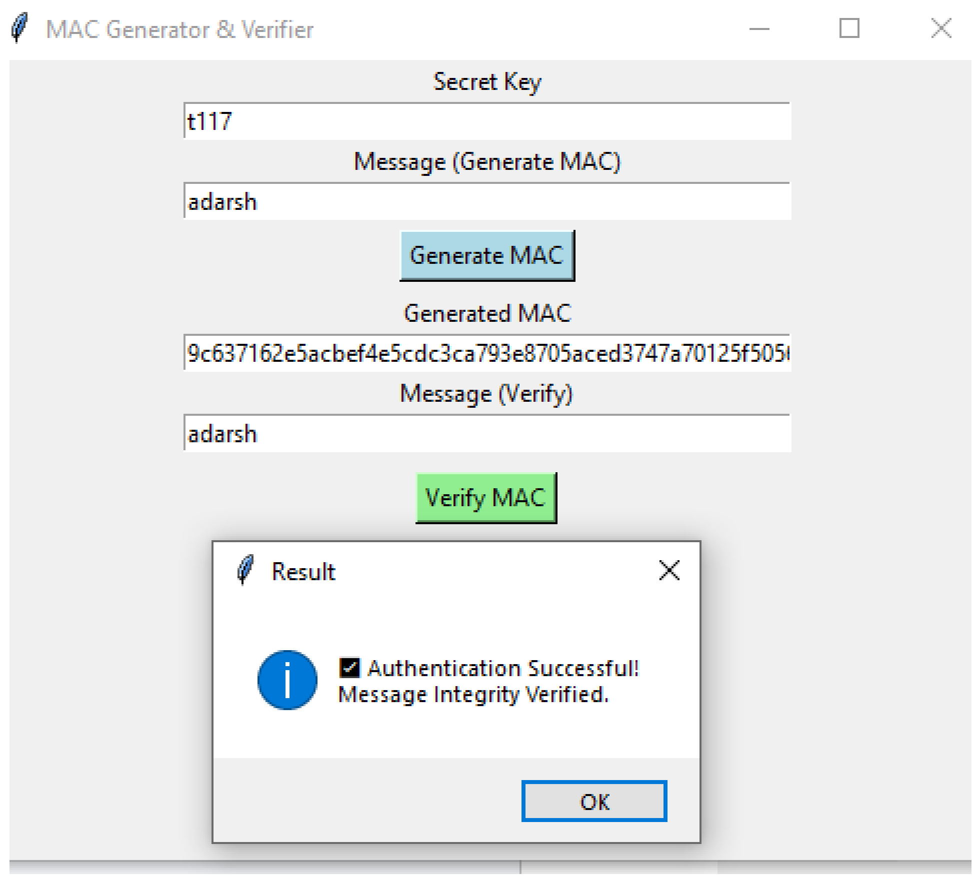

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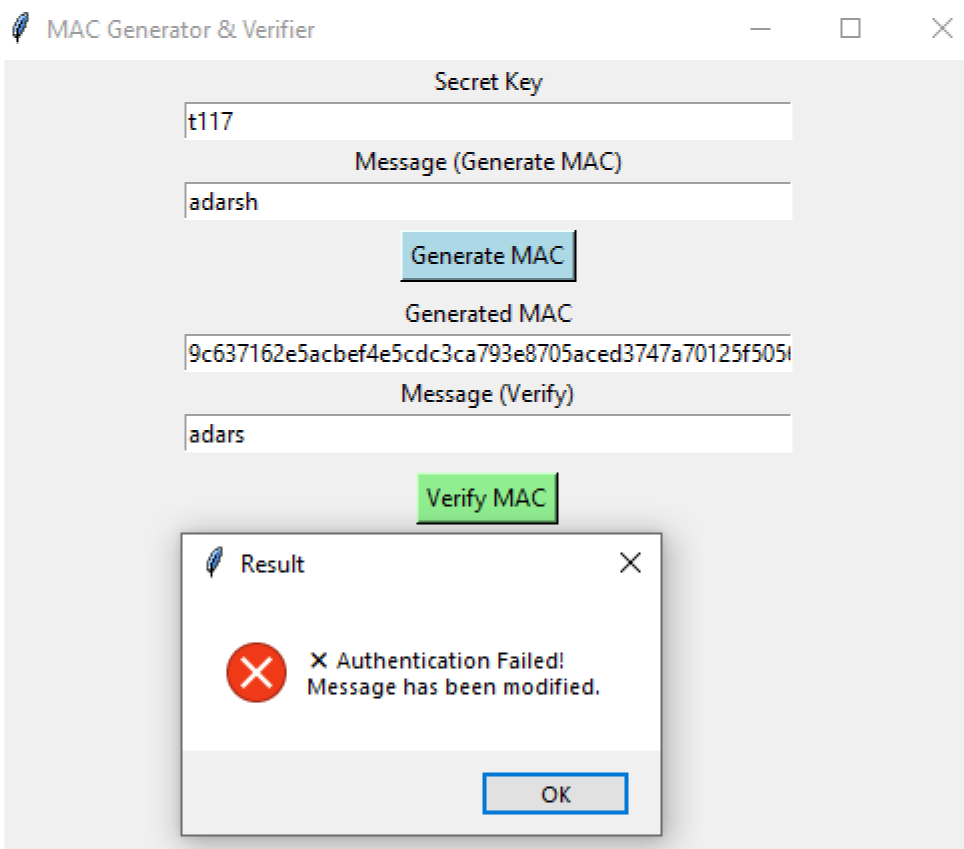

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